



**" IMPLEMENTATION OF END-TO-END ENCRYPTION IN
WHATSAPP AND TELEGRAM APPLICATIONS"**

Written by:

Muhammad Sabil Aditia

Leviyanda Salsabila

Class:

4 SE 2

Faculty :

MR.Feri Nugroho, S.ST., M. IT

Continuing Education Program Center for Computing and Information
Technology

Faculty of Engineering University of Indonesia

2025

PREFACE

Thank you, the author wishes to Allah SWT, the Almighty God for His blessings and mercy, that the author can complete this ISAS task, both in the form of paper and in the form of presentations in a timely manner. Thank you especially for the faculty, Mr.Feri Nugroho, S.ST., M. IT who always help. Thank you to my fellow students who have supported us, and also thank you for being a partner in this education at CCIT-FTUI. The ISAS Paper entitled “Implementation of End-to-end Encryption in Whatsapp and Telegram Applications” which the authors propose as ISAS's 2025 task requirements.

Hope from the author, hopefully, this paper can be useful for the reader, so that it can add knowledge and insight. The author realizes that this paper is not perfect. Therefore, the authors really expect all constructive suggestions and criticisms from the reader to improve this paper. Finally, hopefully, this paper can provide many benefits for the readers.

Depok, 18 April 2025

Author

TABLE OF CONTENT

PREFACE.....	i
TABLE OF CONTENT.....	ii
TABLE OF FIGURES.....	iii
CHAPTER I INTRODUCTION	4
I.1 Background.....	4 I.3
Problem Domain	4 I.4
Writing Methodology.....	5 I.5
Writing Framework.....	5
CHAPTER II BASIC THEORY	6
II.1 Concepts of security.....	6 II.2
Concepts of encryption	7 II.3
Encryption data types.....	8 II.4
Concepts of End-to-end encryption	9
CHAPTER III PROBLEM ANALYSIS.....	11
III.1 How End-to-end encryption works in WhatsApp.....	11 III.2
How Encryption Works In Telegram.....	13 III.3 End
To end Encryption method used in WhatsApp.....	15 III.4
Comparing Secure Messaging Apps.....	19
CHAPTER IV CONCLUSION AND SUGGESTION	20
IV.1 Conclusion	20 IV.2
Suggestion.....	20
Bibliography.....	21

TABLE OF FIGURES

Figure 2. 1 Security principles.....	6
Figure 2. 2 different encryption methods.....	8
Figure 3. 1 encryption process.....	11
Figure 3. 2 End-to-end encryption layer.....	12
Figure 3. 3 Overview of how simple works.....	12
Figure 3. 4 Overview of how simple works.....	13
Figure 3. 5 Payload of a single message to be encrypted in the regular chat.....	14
Figure 3. 6 The AES key and IV is derived from the auth_key and SHA-1 hash of the payload msg_key.....	14
Figure 3. 7 Exchanging messages process.....	18
Figure 3. 8 Message formation.....	18
Figure 3. 9 Comparing secure messaging apps	19

CHAPTER I

INTRODUCTION

I.1 Background

End-to-End Encryption (E2EE) is an important technology in maintaining the privacy and security of digital communications, especially in apps like WhatsApp and Telegram. E2EE ensures that only the sender and receiver can read or access the contents of messages, voice calls, or videos. Without E2EE, private conversations can easily be intercepted by unauthorized parties.

As the use of apps to share private and sensitive information increases, so does the need for secure communication. WhatsApp and Telegram have addressed this need by implementing E2EE. WhatsApp uses Signal Protocol, which provides comprehensive encryption for all types of communication. Telegram uses MTProto, which provides E2EE only for Secret Chats, while regular chats are only encrypted between the server and the client.

E2EE implementation on WhatsApp and Telegram:

- WhatsApp has fully implemented E2EE using Signal Protocol, an encryption system known to be highly secure. All communications on WhatsApp-whether text messages, voice calls, video calls, or files-are protected with E2EE by default.
- Telegram has a different approach. It uses a homegrown encryption system called MTProto. However, E2EE is only available on Secret Chats. For regular (non-secret) chats, Telegram only uses encryption between the user's device and their server (server-client encryption), not end-to-end. This means that Telegram can still technically access the contents of regular chats.

This chapter discusses the importance of E2EE in maintaining the confidentiality of digital communications, as well as how this technology is adapting to the times and users' needs for data privacy.

I.2 Writing Objective

The primary objective of this paper is to provide an in-depth understanding of End-to-end Encryption in the context of WhatsApp and Telegram, elucidating its purpose and functionality.

I.3 Problem Domain

This paper aims to explore the concept, components, workings, advantages, and potential challenges associated with End-to-end Encryption in the WhatsApp and Telegram application.

I.4 Writing Methodology

The writing method is an observation method. That is the collection of materials that will be discussed, analyzed, and examined from official resources and also use “learn and do” which is very influential in research, and draws conclusions from all reports obtained from research.

I.5 Writing Framework

To make it easier the writing the ISAS, Then the discussion be organized with the system of writing.

1.5.1 Chapter I: Introduction

In this chapter 1, it will talk about the background of the problem, writing objective, problem domain, the method of writing that used is the system of writing, and writing framework.

1.5.2 Chapter II: Basic Theory

This chapter describes the concept of security, concepts of encryption, encryption data types, and the principle of end-to-end encryption.

1.5.3 Chapter III: Problem Analysis

This chapter has a more in-depth discussion of encryption End-to-end. Also, this chapter discusses how End-to-end works in WhatsApp and Telegram, the method used, and Comparing Secure Messaging Apps.

1.5.4 Chapter IV: Conclusion and Suggestions

In this chapter contains the conclusions of the results of writing and Suggestions.

CHAPTER II

BASIC THEORY

II.1 Concepts of security

Security means protecting something from harm or danger, whether that harm is done on purpose or by accident. For example, national security protects a country's land, people, and resources from threats. In the same way, organizations also need protection. They must keep their buildings, systems, employees, and especially their information safe. In the world of technology, security often refers to **information security**, which is the protection of data and computer systems from people who are not allowed to access or change them. The goal is to keep information private (confidentiality), accurate and unchanged (integrity), and available whenever it's needed (availability). To achieve this, organizations use different security methods and tools to reduce the risk of any kind of attack or loss.



Figure 2. 1 Security principles

Security, inherent to the protection of information and information systems, encompasses several key principles that guide its implementation and effectiveness:

1) Confidentiality

Confidentiality focuses on preserving the secrecy and privacy of information. It ensures that sensitive data is accessible only to authorized individuals and protected from unauthorized access, disclosure, or interception. Measures such as encryption, access controls, and secure communication channels are implemented to maintain confidentiality. By safeguarding privacy, confidentiality prevents unauthorized disclosure of sensitive information, trade secrets, or personal data, thereby protecting individuals' rights and maintaining trust.

2) Integrity

Integrity ensures that information remains accurate, consistent, and trustworthy. It involves protecting the data from unauthorized modification, deletion, or tampering, both in transit and at rest. Measures such as data validation, checksums, digital signatures, and secure storage mechanisms are employed to maintain integrity. By ensuring data remains unaltered and reliable, integrity safeguards against intentional or unintentional modifications, data corruption, or the insertion of malicious code. It helps maintain the accuracy and reliability of information, which is crucial for decision making, accountability, and maintaining trust.

3) Availability

Availability ensures that authorized users have timely and uninterrupted access to information and system resources when needed. It involves ensuring that information systems and services are operational, responsive, and accessible, even in the face of unexpected events or attacks. Measures such as redundancy, fault tolerance, disaster recovery planning, and robust network infrastructure are implemented to maintain availability. By minimizing disruptions, downtime, or service outages, availability ensures that critical operations can continue, supporting business continuity, productivity, and user satisfaction.

Security is crucially important because it ensures the protection of sensitive information, systems, and assets from various threats and risks. Confidentiality, integrity, and availability, the core principles of security, address key aspects of protecting information. In today's interconnected and digital world, security threats are increasingly sophisticated and prevalent. Cyberattacks, data breaches, and unauthorized access pose significant risks to organizations, their stakeholders, and individuals. Security measures help mitigate these risks by implementing controls, safeguards, and countermeasures to protect against potential threats. Additionally, security is vital for regulatory compliance. Many industries and jurisdictions have specific legal and regulatory requirements regarding the protection of sensitive information. Failure to meet these requirements can result in severe financial and reputational consequences.

II.2 Concepts of encryption

Encryption is a way to protect information by turning it into a secret code. It helps make sure that only the right people can read or understand the message. For example, when you send a message to someone, encryption changes it into something that looks like random letters and numbers. This way, if someone else tries to read it, they won't understand it.

Before encryption, the message is called **plaintext**—it's easy to read. After encryption, it becomes **ciphertext**, which looks like nonsense unless you have the **key**. The key is like a special password or code that helps unlock and read the real message.

Encryption is used in many parts of our daily lives. It protects your messages on apps like WhatsApp, your passwords when you log into websites, and your credit card information when you shop online. Without encryption, anyone could steal or read your private data.

There are two main ways encryption protects your data. The first is **data at rest**, which means the information stored on your phone, computer, or in the cloud. The second is **data in transit**, which is information being sent over the internet, like when

you send a message or upload a file. Encryption keeps your data safe in both cases.

Encryption uses a math formula called an **algorithm**. This formula works with a **key** to lock and unlock the message. There are two types of encryption: **symmetric** and **asymmetric**. In symmetric encryption, the same key is used to lock and unlock the data. In asymmetric encryption, there are two keys: a **public key** to lock the message and a **private key** to unlock it. This type is often used in online banking and secure websites.

II.3 Encryption data types

Encryption uses a special code called a **key** to lock (encrypt) and unlock (decrypt) information. This key is made using a formula called an algorithm and is always random and unique. There are two main types of encryption: **symmetric** and **asymmetric** encryption.

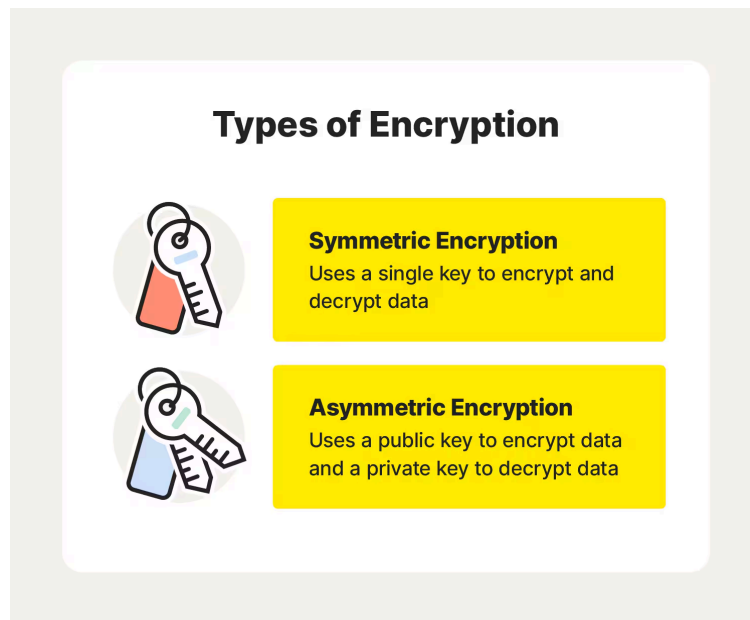


Figure 2. 2 different encryption methods

1) Symmetric encryption

Symmetric encryption uses **only one key** to both encrypt and decrypt the data. This means the same key must be shared between the sender and the receiver. The key can be a password, a code, or a random string of letters and numbers. This type of encryption is often used in things like banking systems because it is simple and fast. Symmetric encryption comes in two forms: **block algorithms**, which encrypt chunks of data at once, and **stream algorithms**, which encrypt data one piece at a time. Although it has some weaknesses—like the risk of the key being stolen—it is still popular because of its speed and efficiency.

2) Asymmetric encryption

Asymmetric encryption, also called **public key encryption**, uses **two different keys**—a public key and a private key. The public key is shared with others and can be used to encrypt or decrypt data. The private key is kept secret and is only known by the owner. These keys work together, depending on the goal. If someone

- **encrypts data using a public key**, only the person with the matching private key can read it. This keeps the message safe, even if someone else tries to intercept it. On the other hand, if data is
- **encrypted using a private key**, the receiver can use the public key to check that the message really came from the right sender. This helps prove the sender's identity and ensures the message hasn't been changed.

Because asymmetric encryption uses two longer keys, it's much slower and less efficient to run compared to symmetric cryptography. It can even bog down networks and create issues with memory capacity and battery life. However, asymmetric encryption is considered more advanced in terms of security than symmetric cryptography. Both are still in use today—sometimes simultaneously to compensate for the other's weaknesses.

II.4 Concepts of End-to-end encryption

End-to-end encryption, or E2EE, is a way to keep messages private and secure when they are sent from one person to another. It works by turning the message into a secret code on the sender's device, and only the receiver's device can turn it back into a readable message. This means that even while the message is traveling through different servers or networks, no one else—not even the internet provider or the app company—can read what it says.

Encryption itself is the process of using a special formula, called an algorithm, to change normal text into something unreadable. This process also uses something called an **encryption key**, which works like a digital password that scrambles and unscrambles the message. Only someone with the correct key can read the message.

End-to-end encryption uses this same method, but it adds an extra layer of protection by making sure the data stays private from the moment it is sent until the moment it is received. However, E2EE only protects the actual **content** of the message. It does not hide other information, like who sent the message, who received it, or the time it was sent. This extra information is called **metadata**, and it can still be seen by service providers or others.

In short, end-to-end encryption helps keep your messages safe and private by making sure only you and the person you're communicating with can read them.

So, do whatsapp and telegram apps use encryption?

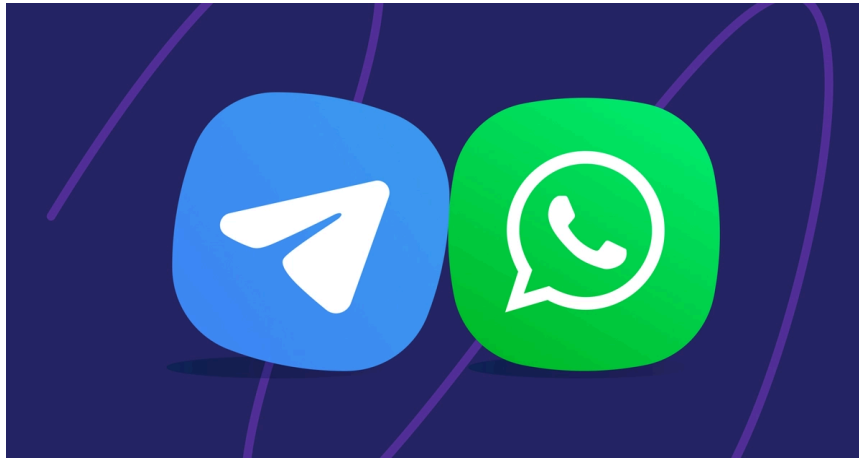


Figure 2. 3 encryption in whatsapp and telegram

Both messaging apps are very popular in Indonesia and widely used for personal and business communication. WhatsApp and Telegram employ end-to-end encryption to protect the content of messages exchanged between users. WhatsApp uses the Signal Protocol developed by Open Whisper Systems, which implements end-to-end encryption for all types of communication, including text messages, voice calls, video calls, and file sharing. This means that the messages are encrypted on the sender's device and can only be decrypted by the recipient's device, ensuring that no third party, including WhatsApp itself, can access the content of the messages. Telegram, on the other hand, uses a proprietary encryption protocol called MTProto. While Telegram offers end-to-end encryption for its "Secret Chats" feature, regular chats are not end-to-end encrypted by default. Instead, they are encrypted using server client encryption, where the messages are encrypted between the client and Telegram's servers, but Telegram can access the content of the messages on their servers.

It's important to note that while end-to-end encryption provides strong privacy protection for the content of messages, it does not protect metadata such as the sender's and recipient's

identities, the time of communication, or the IP addresses involved. This metadata can still be accessed by the service providers or intercepted by third parties.

CHAPTER III

PROBLEM ANALYSIS

III.1 How End-to-end encryption works in WhatsApp

Cryptography is a method used to protect information and communication channels using specific codes that can only be accessed by authorized users. One of these methods is encryption, where open messages (plaintext) are transformed into secret random codes, making the actual information inaccessible. This process is known as encryption and decryption. The end-to-end encryption method in the WhatsApp application ensures that only the user and the intended recipient can access the communication. It is secured with a key, and only the recipient who knows the key can decrypt the message, making it impossible for any intruder, including the service provider, to do so.

WhatsApp's end-to-end encryption mechanism operates automatically without requiring specific user settings. Messages are encrypted before being sent to the Base Transceiver Station (BTS), which serves as a telecommunication infrastructure facilitating wireless communication between communication devices and the operator's network. For additional security, each message sent has a unique key. Here's an example of the encryption process:

	H	E	L	L	O
ASCII	072	069	076	076	079
					+1(kunci)
Dienkripsi pesan	072	070	077	077	080
					I F M M P
					-1(kunci)
Dideskripsi	072	069	076	076	079

Figure 3. 1 encryption process

(REF:<https://journals.penerbitjurnal.com/index.php/teknik/article/view/14/14>)

With this system, the data distribution system is highly secure. Each message sent has a unique key used to encrypt the message. Every chat possesses a security code to verify encrypted end-to-end calls or chats. The fundamental concept of end-to-end encryption is as follows:

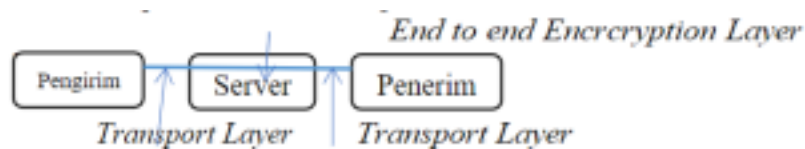


Figure 3. 2 End-to-end encryption layer

(REF:<https://journals.penerbitjurnal.com/index.php/teknik/article/view/14/14>)

In the illustration, it appears as if end-to-end encryption has a separate pathway for communication, but this pathway is merely metaphorical, serving as a wrapper for the data. The end-to-end encryption feature in the WhatsApp application is designed to thwart third-party attacks attempting to pilfer information, be it messages, images, videos, photos, documents, voice messages, status updates, or calls. Even if a hacker manages to steal the encryption key, they would still be unable to decrypt the sent messages.

```

PS C:\Users\lgd1\Downloads\e2e-encryption-demo> python client.py s
ugeni

secret_key=13
public_key=21

system >> Penunggu teman chat...
system >> Ready! (e2e key=6)
HALO
Tarno >> lagi apa sugeni

PS C:\Users\lgd1\Downloads\e2e-encryption-demo> python client.py
Tarno

secret_key=10
public_key=12

system >> Ready! (e2e key=6)
sugeni >> HALO
lagi apa sugeni
  
```

Figure 3. 3 Overview of how simple works

Pictured above is a simple chat application that ensures messages sent between users remain secure and encrypted. Communication between users occurs over a network connection using TCP sockets, and every message sent is encrypted so that it cannot be read by unauthorized parties. The program also allows users to exit the chat securely using the "quit" command. With these simple features, two users can communicate through the terminal in a secure and private manner. The process is executed simultaneously with the help of threading, allowing users to receive and display messages instantly. Overall, the program provides a secure and easy-to-use chat experience.



```
PS C:\Users\Agil Andhya\Downloads\e2e-encryption-demo-main 22\e2e-encryption-demo-main> python server.py
[INFO] Server berjalan di localhost:8000
{"type": "init", "name": "Sugeno", "pubkey": 8}
{"type": "init", "name": "Tama", "pubkey": 8}
{"type": "message", "text": "P6RlpD8q2a8LCrzkFyqHA/c6kIb+CTWSP9x/vLEBjsa"}
{"type": "message", "text": "Opug58298ejIu8Dw8ILDU8u84u8hlcIjKer9Dzr575Me"}
{"type": "message", "text": "gPFe8b8Gudaal/5UqIVuywond8g5SPZ7B0CN2zrtp24e"}
```

Figure 3. 4 Overview of how simple works

The image above is a sort of "call center" for chat apps that ensures messages sent and received by users remain secure and unreadable by others. Every time someone enters a chat room, they let the manager (server) know that they are there. The server records their information and lets them know if there is someone else to chat with. When there are two people ready to talk, the server tells them how to communicate securely using a secret key that only they both know.

All messages sent by these people are encrypted before they reach their chat buddy. So, even if the message crosses the internet network, it remains secure. If someone decides to leave the chat, the server deletes their information and ensures that no one can access the chat afterwards. The program is designed to handle two people who want to talk securely and provide a way for them to communicate without worrying about their messages being seen by others.

III.2 How Encryption Works In Telegram

Initialization During the first launch of the Telegram application, the user needs to enter and verify their telephone number. The verification is done by sending a five digit code to the phone via SMS. The user then enters the code into the app and by doing so verifies the phone number. When this process is done, the registration process begins. The gist of the registration process is to create a secret key called `auth_key` which is then used for all client-server communication and regular chats. Next, a fingerprint of this key, labeled as `auth_key_id`, is created. It is crafted from the last 64 bits of `SHA1(auth_key)`. We are omitting the details here, however we reference the full analysis report should the reader be inclined to find out more.

Payload Let us describe the content of a single message further referred to as a payload. The payload prior to encryption is depicted in Figure 1 and it contains:

1. salt Periodically changed number used for various protection purposes.
2. session_id Unique number to identify the user and their device.
3. msg_id Unique ID of the message within a session.
4. seq_no Message sequence counter.
5. length Length of the actual message.
6. message Content of the actual message.

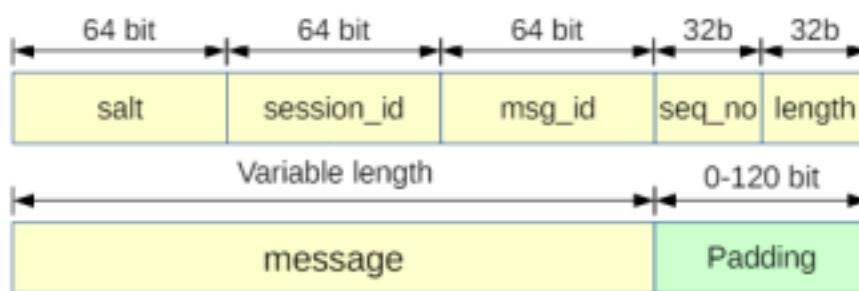


Figure 3. 5 Payload of a single message to be encrypted in the regular chat

(<https://dl.acm.org/doi/abs/10.1145/3150376.3150382>)

Payload of a single message to be encrypted in the regular chat contains salt, session and message identifier, sequence number, length and the text itself. First, the message key `msg_key` is calculated. It is composed of the 128 least significant bits of the SHA-1 hash of the payload to be encrypted. Next, the array is padded with 0-120 random bits in order to be divisible by the AES block size – 128 bits.

This `msg_key` along with the `auth_key` obtained during the initialization process are taken as input for the Key Derivation Function (KDF) which performs a number of SHA-1 hashes and truncations, yielding two 256-bit values: the AES key and the IGE initialization vector (IV) used for encrypting this particular message.

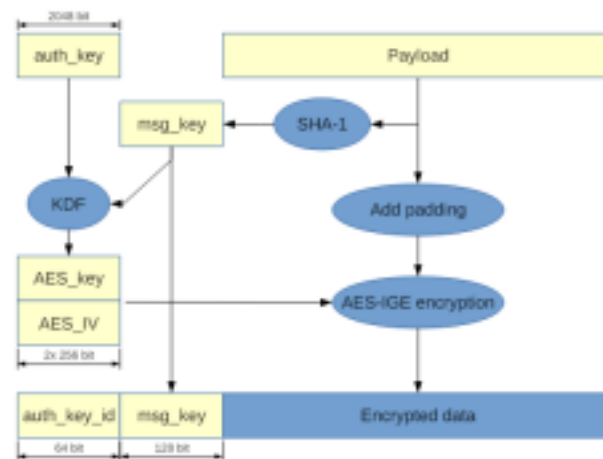


Figure 3. 6 The AES key and IV is derived from the `auth_key` and SHA-1 hash of the payload `msg_key`.

(<https://dl.acm.org/doi/abs/10.1145/3150376.3150382>)

The AES key and IV is derived from the `auth_key` and SHA-1 hash of the payload `msg_key`. Using these values the padded payload is encrypted. The encrypted data along with `auth_key_id` and `msg_key` are ready to be transported. Finally, two other values are added on top of the encrypted data array: the already mentioned `msg_key` and the `auth_key_id`. The data array is then ready to be transported to the server.

III.3 End To end Encryption method used in WhatsApp

WhatsApp uses a strong end-to-end encryption method to protect the privacy and security of users' messages, including:

A. Public Key Types

- 1) IdentityKeyPair is a Curve25519 medium-term key pair
- 2) SignedPreKey is a Curve25519 medium-term key pair, generated during the application installation process, and will change over time.
- 3) One-Time Pre Keys are rows of Curve25519 key pairs that are used only once, generated during the installation process, and will appear when needed.

Some of the public key types used for end-to-end encryption on the WhatsApp Messenger App, are Curve25519 type key pairs. Curve25519 is an elliptical curve (ecc) with a length of 128 bit that uses the ECDH (elliptic curve Diffie-Hellman) concept. The good security and some advantages compared to other key exchange algorithms are considerations for the use of Curve25519. In Daniel J. Bernstein's paper entitled "Curve25519: new Diffie-Hellman speed records" it is recognized that Curve25519 has good computational speed that can be applied to cryptographic algorithms.

This is what might underlie WhatsApp Messaging using curve25519 in the end-to-end encryption feature. As we know, every mobile application developer in making applications must be more efficient, especially from memory usage which can affect the efficient running of the application.

To authenticate and encrypt messages between 2 users, the basic concept used in Curve25519 is that each user has a 32 Byte secret key and 32 Byte public key and each pair of users has a 32 Byte shared key.

The steps in the process of generating a shared secret key that will be used as the key to encrypt the message as follows:

- 1) Both users should jointly determine the public string, in the picture above the public string is valued at 9. Then each user determines the secretkey. Secret key is

classified

- 2) Then generate the public key by entering the value of the secret key (a and b) and the value of the public string (a, 9) into the Curve25519 function for the sending user and for the receiving user Curve25519 (b,9).
- 3) Next, the sender and receiver exchange the public keys that have been entered into the Curve25519 function.
- 4) And to calculate, by entering the obtained public key and secret key values into Curve25519.

Some of the advantages that the concept of Curve25519 has are as follows: 1)

ExtremelyHighSpeed

In a scientific work, it has been proven in several Personal Computers (PCs) that Curve25519 has the speed that can be used for a process.

- 2) No Time Variability

Research has been conducted on Curve25519 and the results of the research prove that Curve25519 is resistant to timingattack, hyperthreadingattack, and cachetimingattack.

- 3) Short Secret Key

Secret Key owned by Curve25519 is 32Bytes.

- 4) The Short Public Key

On Curve25519 has a 32Byte public key.

B. Session Key Types

- 1) Root Key is a value whose length is 32 Bytes and functions to generate ChainKeys.
- 2) Chain Key is a value whose length is 32 Bytes and functions to generate MessageKeys.
- 3) Message Key is a value whose length is 80 Bytes serves to encrypt the message and out of 80Bytes, the one used for the AES-256 key is 32Bytes. Out of 80Bytes, 32Bytes are used as the AES-256 key.

C. User Registration (ClientRegistration)

At the beginning of registration, the user sends a Public Identity Key, Signed Pre Key (along with its signature), and a number of PublicOne-Time Pre Key to the application server. The public key and the user's identity are stored by the WhatsApp Messenger server.

D. Initialization of Session Establishment

In order to communicate with other users, the user must set up a session for secure (encrypted) communication. Session Establishment Initialization In order to communicate with other users, the user must set up a session for secure (encrypted) communication. A session is only formed once and if you want to continue communicating with the same user, there is no need to create a new session.

However, if the session that has been established is down due to reinstallation of the application or moving the user to another device, a new session will be created. The process of establishing a session is as follows:

- 1) The user who requests communication with another user is called the Initiator. The Initiator then requests a public Identity Key, a public Signed Key, and a single public One-TimePreKey belonging to another user (recipient).
- 2) The server returns the request with a public key value. One One-TimePreKey can only be used once, so after it is sent and received by the initiator, the One TimePreKey is automatically deleted from the server's storage. The Identity Key belonging to the recipient as the Recipient, the SignedPreKey as the Recipient, and the One-TimePreKey as the Recipient are stored by the Initiator.
- 3) Then the initiator generates an ephemeral Curve25519 key pair as the Initiator.
- 4) Initiator initiates the IdentityKeys with I.
- 5) Next, the initiator calculates the master secret which is $\text{master_secret} = \text{ECDH}(\text{Iinitiator}, \text{Srecipient}) || \text{ECDH}(\text{Einitiator}, \text{Irecipient}) || \text{ECDH}(\text{Einitiator}, \text{Sr ecipient}) || \text{ECDH}(\text{Einitiator}, \text{Orecipient})$.
- 6) And to generate the RootKeys and Chainkeys from the master_secret, the Initiator

usesHKDF.

E. Receiving Session Setup

After establishing the encryption session, the initiator can immediately start sending messages to the receiver, even if the receiver is offline. The initiator establishes corresponding sessions until a response is provided by the recipient. The recipient of the message will calculate the master_secret using the private key and public key attached to the header of the received information packet. Then the message receiver removes the One-Time Pre Key used by the initiator. Then the initiator uses HKDF to get Root Keys and Chain Keys from the master_secret.

F. Exchanging Messages

The first thing to do in communication is to set up a session, after the session is well established, message exchange can be carried out. Messages exchanged between clients on the WhatsApp Messaging Application are protected by Message Keys using AES256 encryption in CBC (Chiper Block Chining) mode and HMAC SHA256 is used as data integrity. Each packet sent, the Message Key is always different because it is ephemeral (short use).Message Key cannot re-encrypt messages.Message Key is obtained from the Chain key initiator and will bounce continuously and return to a certain point "ratchets method".Block diagram of the encryption process is as follows:

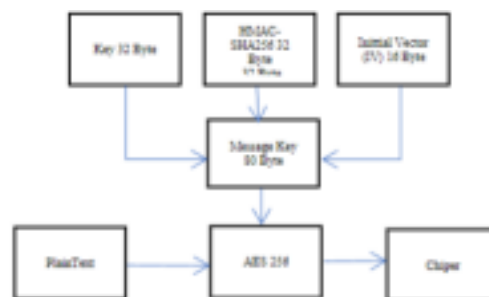


Figure 3. 7 Exchanging messages process

(REF:<https://journals.penerbitjurnal.com/index.php/teknik/article/view/14/14>)

In the figure above, the encryption algorithm used for the key (Message Key) of AES-256 which is 80 Bytes long consists of a 32 Byte key, HMAC SHA256 32 Bytes, and an initial vector (iv) of 16Bytes. In the process of generating the Message Key obtained from the Chainkey (32Bytes). To encrypt data, the initiator needs Message Keys every time. Block diagram of the formation of Messages as follows



Figure 3. 8 Message formation

(REF:<https://journals.penerbitjurnal.com/index.php/teknik/article/view/14/14>)

The way to get the value of the MessageKey is as follows:

- 1) MessageKey=HMAC-SHA256(ChainKey,0x01)
- 2) ChainKey is then updated by the calculation process ChainKey=HMAC SHA256(ChainKey,0x02).

In the message encryption performed during communication, the key (MessageKey) of the generated Chain Key is 32byte in nature "ratchets". The ratchets method was created to overcome the problem of exchanging the same encryption public key and to ensure that the transmitted public keys are different. This feature is a modern security protocol to prevent sniffing attackers from sniffing the network, who will try to perform descriptive analysis while using only one key. The ephemeral key exchange method also makes it difficult for an attacker to obtain the next key because the key is stored in memory only temporarily and for a short period of time.

WhatsApp Messenger's end-to-end encryption feature is also present in the call mode. The communication path used has implemented the SRTP (SecureReal TimeTransportProtocol) method, which is specifically designed for real-time voice or video transmission. This protocol maintains data confidentiality, message

authenticity, and feedback protection from RTP traffic. Root Process users in making calls through the WhatsApp Messenger Application are as follows:

- 1) If a session has not been formed, the Initiator forms a session.
- 2) Then a random 32Byte SRTP masters secret value is generated by the Initiator.
- 3) Then the message sent by the Initiator is encrypted to the recipient in the form of an incoming call sign containing the SRTP masters secret and automatically SRTP encrypts the call.

III.4 Comparing Secure Messaging Apps

Each of the four mentioned messaging apps – Telegram, Signal, Wickr, and WhatsApp is considered a secure online communication option that gives users more privacy. Take a look at the table to compare each of the secure messaging apps:

	Telegram	Signal	Wickr	WhatsApp
End-to-end encryption	Secret chats only	Default	Default	Default
Encryption protocol	Proprietary	Open-source	Proprietary	Open-source
Data collection	Moderate	Minimal	Minimal	Extensive
Self-destructive messages	Only in Secret chats	Possible in all messages	Possible in all messages	Possible in all messages
Geolocation data	Only voluntary sharing	Only voluntary sharing	Only voluntary sharing	Only voluntary sharing

Figure 3. 9 Comparing secure messaging apps

CHAPTER IV

CONCLUSION AND SUGGESTION

IV.1 Conclusion

In summary, this paper discussed how end-to-end encryption (E2EE) is used in popular messaging apps like WhatsApp and Telegram to keep user data safe. Security means protecting information from being accessed or changed by people who shouldn't have access. One important way to do this is through **encryption**, which turns readable data into a secret code that only someone with the correct key can read.

End-to-end encryption is a special type of encryption where only the sender and the intended receiver can read the message. Even the app itself or internet providers can't access the message content. This kind of encryption helps keep private conversations truly private and protects sensitive information from being stolen or misused.

Overall, E2EE improves user privacy and security, builds trust in digital communication, and reduces the risk of data leaks. It's a powerful tool to ensure that messages stay between the people they are meant for.

IV.2 Suggestion

In this paper, the importance of end-to-end encryption in preserving privacy and security in digital communication has been discussed. End-to-end encryption ensures that messages can only be read by the intended senders and recipients, while protecting them from unauthorized access by third parties. By adopting end-to-end encryption, users can feel safer and more protected when sharing sensitive information.

Bibliography

Jeeun Lee, C. (2017). Security Analysis of End-to-End Encryption in Telegram. *SCIS*, 24-27.

Michael Bolli, P. K. (2017). *WhatsApp End-to-End Encryption*. Bern University of Applied Sciences.

Michael E. Whitman, H. J. (2017). *Principles of Information Security*. USA: Cengage Learning.

Salomon, D. (2005). *Foundation of Computer Security*. Germany: Springer Science & Business Media.

Siti Khasanah, T. S. (2023). ANALISIS PENCEGAHAN PENCURIAN DATA MELALUI APLIKASI WHATSAPP MENGGUNAKAN METODE KRIPTOGRAFI. *Jurnal Sain dan Teknik*, 145-153.

Sušánka, K. (2017). Security Analysis of the Telegram IM. *ROOTS*, 16-17.

[https://securitymadesimple.org/cybersecurity-blog/what-are-the-3-principles-of-information security/](https://securitymadesimple.org/cybersecurity-blog/what-are-the-3-principles-of-information-security/) [16/03/2024]

<https://www.tutorialspoint.com/what-is-encryption-basic-concepts-process-and-types> [16/03/2024]

<https://www.ibm.com/topics/end-to-end-encryption> [16/03/2024]

<https://nordvpn.com/blog/what-is-end-to-end-encryption/> [16/03/2024]

<https://www.techtarget.com/searchsecurity/definition/end-to-end-encryption-E2EE> [17/03/2024]

<https://blog.etesync.com/end-to-end-encryption-what-it-is-and-why-it-is-needed/> [17/03/2024]

<https://nordvpn.com/id/blog/is-telegram-safe/#:~:text=the%20secret%20chats.-,Does%20Telegram%20provide%20end%2Dto%2Dend%20encryption%3F,device%20instead%20of%20Telegram's%20server> [01/06/2024]

<https://core.telegram.org/api/end-to-end> [01/06/2024]

